

AI server might contain eight or more accelerator cards, each with its own power delivery system and generating its own EMI profile. Multiply this across a rack of ten such servers, then across a row of twenty racks, and the result is an electromagnetic environment that resembles a poorly managed radio spectrum in a crowded city.

Shielding, which was adequate in previous generations of compute infrastructure, is struggling to keep pace. Traditional server chassis were designed around the EMC requirements of their era. The field strengths now generated inside dense AI racks exceed what those designs anticipated, and the frequencies involved are sufficiently high that even small gaps in shielding, ventilation slots, cable cut-outs, and connector openings begin to act as antennas.

The wireless sensor problem

One of the less discussed consequences of this situation involves the environmental monitoring infrastructure on which modern data centres depend. Temperature sensors, humidity sensors, and power monitoring nodes are increasingly wireless in newer facilities because cable management at scale is an engineering and operational nightmare. These sensors typically operate in the 2.4GHz or 868MHz bands, frequencies that sit squarely within the EMI emissions generated by high-density AI compute hardware.

In several large-scale AI training facilities, operators have reported intermittent sensor drop-outs that correlate closely with peak training workloads. The data centre's

environmental awareness, the very system designed to prevent thermal runaway, becomes unreliable precisely when thermal loads are at their highest. This is not a theoretical risk. It has caused operational disruptions and, in at least a few documented cases, contributed to thermal events that required emergency workload shedding.

Cooling system interference

The interference problem extends to cooling control systems as well. Precision liquid cooling, now common in AI data centres because of the heat densities involved, relies on sensors and controllers that must communicate reliably in real time. Variable-speed pump controllers and intelligent coolant distribution units use power electronics that themselves generate EMI, creating a situation in which the solution to one problem becomes the source of another.

Researchers at several European data centre operators have begun instrumenting their facilities with spectrum analysers specifically to characterise the EMI environment at different workload levels. The preliminary findings are striking: the noise floor in a fully loaded AI training rack can be 30dB to 40dB higher than in an equivalent compute rack from five years ago. At those levels, conventional wireless sensor protocols simply cannot maintain reliable links.

What the industry needs to do

The path forward requires action on several fronts simultaneously. At the hardware level, AI accelerator designers need to treat EMI as a

first-class design constraint rather than an afterthought addressed only during the final PCB review. This means adopting more sophisticated power delivery architectures featuring spread-spectrum switching, improved on-chip decoupling, and chassis designs that treat high-frequency shielding as seriously as thermal management.

At the system level, data centre operators need to invest in characterising the EMI environment of their facilities, not merely at commissioning but continuously as workload patterns evolve. Spectrum-monitoring infrastructure, once reserved for wireless network management, needs to become standard in AI data centres. The cost is trivial compared with the cost of an undetected thermal event.

At the protocol level, wireless sensor systems deployed in AI compute environments need to use frequency hopping, adaptive power control, and robust error correction, capabilities available in standards such as WirelessHART and ISA100. But these are not generally deployed in data centre monitoring applications because the interference environment was previously not severe enough to require them.

The AI data centres are rapidly becoming one of the most electromagnetically hostile environments in which electronic systems must operate. The engineering community that built the radios, sensors, and compute hardware that created this situation is also best placed to solve it. That work needs to begin now, before the problem scales faster than the available solutions can keep up. **EFY**

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